

Graphs 6

Review

1 $y = x^2 - 2x - 1$

$y' = 2x + 2x^{-2}$

At stationary point, $y = 0$

$2x + \frac{2}{x^2} = 0$
 $x^3 = -1$

$x = -1$

$y = (-1)^2 + \frac{2}{-1}$
 $= 3$

min. point at $(-1, 3)$ ✓

$y'' = 2 - \frac{4}{x^3}$

$y''(-1) = 2$

∴ point is a minimum ✓

2 c) $C = 6 + 4(b-1)$ ✓

3 $x^2 \tan x + \cos 2x - 3 \sin 4x$

$x^2 + (-\frac{1}{2}\cos^2) - 12x$

$x^2 + 1 - 2x^2 - 12x$

$-x^2 - 12x + 1$

$c = 1$

$b = -12$ ✓

$a = 1$ ✓

4 d) ✓

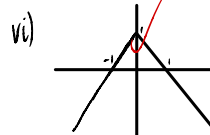
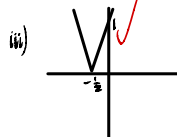
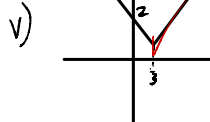
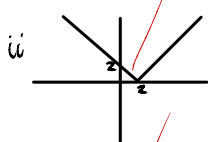
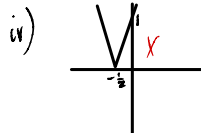
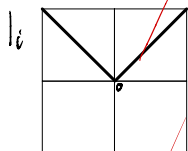
5 $(x-1)^2 - 1 + (y+3)^2 - 9 = 10$

$(x-1) + (y+3)^2 = 20$

$r = \sqrt{20}$ ✓

$C = (1, -3)$ ✓

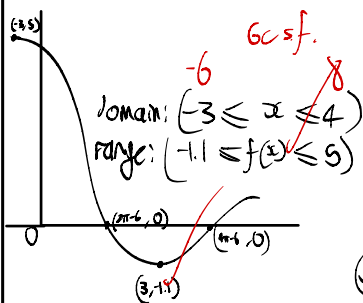
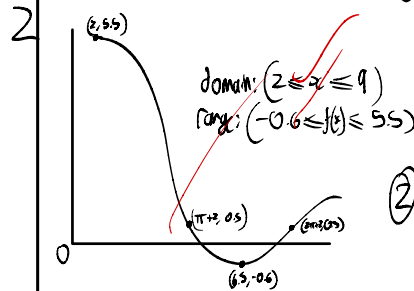
Consolidation



5 $y = 5x^2 - 15x + 4$

i $y = 5(x - \frac{3}{2})^2 - \frac{45}{4} + 4$
 $= 5(x - \frac{3}{2})^2 - \frac{29}{4}$

translate of $(\frac{3}{2}, -\frac{29}{4})$
stretch sf. $\frac{1}{5}$ parallel to y axis



3 $x^2 - 2x$
 $\rightarrow (x+2)^2 - 2(x+2) + 1$
 $\rightarrow (3x+2)^2 - 2(3x+2) + 1$
 $\rightarrow (-3x+2)^2 - 2(-3x+2) + 1$

i) $9x^2 - 12x + 1 + 6x - 4 + 1$
 $= 9x^2 - 6x + 1$
 $= (3x-1)^2$

ii) line of symmetry: $x = \frac{1}{3}$

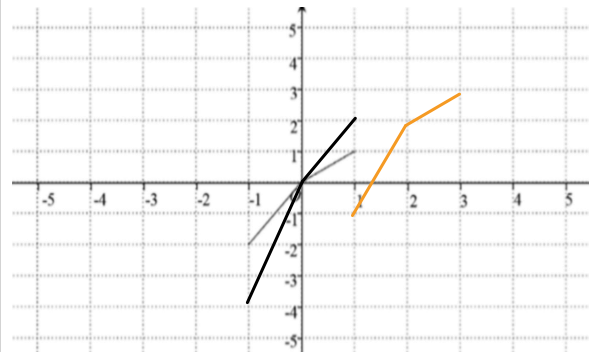
iii) turning point: $(\frac{1}{3}, -\frac{2}{3})$

4 i) translate by $(-3, 0)$

ii) stretch vertically by scale factor $\frac{1}{3}$
stretch horizontally by scale factor 2
reflect in x-axis

Exam Qs

1) Reflect in line $y = x$
stretch by sf. $\frac{1}{3}$ parallel to x axis
translate by $(\frac{2}{3})$



3 a) $y = -\sin(x)$

b) $y = -\sin(x) + 1$

